

Syllabus

BAE5028 Future Mobility: from privately owned Vehicles (ICEVs) to de-carbonised autonomous mobility on demand

Prof. Dr. Guy Fournier
Summer Semester 2026

Level	Master	
Credits	3	
Student Contact Hours	2	
Workload	90 hours	
Prerequisites	none	
Time	s. LSF	
Room	s. LSF	
Start Date	s. LSF	
Lecturer(s)	Name	Prof. Dr. Guy Fournier
	Office	T1.3.29
	Virtual Office	Virtual Office Prof. Fournier
	Office Hours	Monday, 13:45-15:15 individual appointments should be arranged per e-mail (see below)
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Summary

This course aims to advance the student's knowledge in future mobility. Mobility is a key element of our societies but is increasingly criticised because of the external costs it generates. The ques-

tion is how could future mobility be more sustainable in terms of economic, social and environmental impact. Different new technologies and business models will thus be analysed and evaluated.

Main learning goals are to:

- to increase comprehension on how environment and in general external costs in transport like congestion costs, noise, accidents, climate change, air pollution, habitat damage or resource availability influence future mobility,
- to introduce future automotive technologies and mobility concepts and to evaluate them.
- to open student's mind and develop critical thinking skills.

The course is to be assigned to the "Technology" part of the elective courses.

Outline of the Course

During a semester approximately 14 sessions are available to consider the following main topics:

- Drivers of the new mobility paradigm
- Options for the new individual mobility:
 - o Short-term strategies:
 - Aerodynamics,
 - Improved drive resistance,
 - Improved energy efficiency of car components (e.g. powersteering, air conditioning, alternator),
 - Light weight design,
 - Downsizing.
 - o Biofuel and synthetic fuels
 - o Hybrid vehicles
 - o Electric vehicles
- Electric vehicles:
 - o operating modes
 - o new value chain (e.g. electric motor with permanent magnets or electromagnets, power electronics, battery, infrastructure etc.)
- Automated vehicles:
 - o Operating modes and levels of automation,
 - o automated technologies:
 - Sense (perceive relevant static objects, determine the location of AV, predict the future behaviour of relevant actors) with sensors like GNSS, Radar, Camera, LiDAR, V2X)
 - Plan (create a collision free and lawful driving plan)
 - Act (correctly execute and actuate the driving plan), micro- and macro navigation
 - o Fields of application (truck, shuttle, car, train etc.)
 - o Player/market
 - o Legislation; Obstacles
- Mobility and services:
 - o EV as a part of mobility solutions: Mobility as a service (Maas)
 - o EV as a part of storage solutions: V2G, P2G

- Project and simulation reports:
 - o EV and new mobility services and value-added services (V2G)
 - o Car-sharing project in Baden-Württemberg
 - o Robocabs simulation in Berlin and Stuttgart
 - o Automated EV as mobility gap filler in a MaaS (mobility as a Service) and/or an ITS (Intelligent Transport System):
 - “AVENUE” (Horizon 2020: 2018-2022) and
 - “ULTIMO” (Safe, Resilient Transport and Smart Mobility services for passengers and goods: 2022-2026) project (Horizon Europe)
- Industry report: Invited Guest: Martin Lischka, Director Marketing, Brand & Strategy: Sustainable mobility for people and goods at Quantron AG

Course Intended Learning Outcomes and their Contribution to Program Intended Learning Outcomes / Program Goals

Program Intended Learning Outcomes		Course Intended Learning Outcomes	
After completion of the program the students will be able...		After completion of the course the students will be able...	
1	Responsible leadership in organizational contexts		
2	Creative problem solving skills in a complex business environment		
2.1	...to recognize and define problems as well as assess their importance.	... to gain insights into the problem of the change of the powertrain towards electrification and to assess alternatives.	
2.2	...to analyse complex in-company and inter-company problems and challenges from different perspectives and/or within an international context.	... to apply knowledge of technologies and solutions in electromobility and automated mobility.	
2.4	...to clarify successfully complex problems and solutions to both experts and laymen.	... to overview the challenges and technical approaches of electromobility and automated electromobility.	
3	Creative problem solving skills in a complex business environment		
4	Interdisciplinary and integrative work		
4.1	...to apply their expert knowledge in the engineering and business field and to provide integrated solutions to complex tasks.	... to understand the interaction of possible solutions, new services and economic conditions.	

Teaching and Learning Approach

The course consists of 6 sessions (see time table) and awards 3 credits for a successful completion of the course. It is therefore expected that student's workload will amount at least 90 hours to prepare themselves for the sessions, attend and revise the lectures.

The teaching and learning concept is characterised by a three-phase approach, whereby the understanding of interrelationships is clearly in the foreground in comparison to the mere provision of information. In phase I and thus the essential part of the course is taught by the lecturers in the form of a lecture of the relevant knowledge. Here, the students are introduced to the relevant knowledge through e.g. questions and discussions and are actively involved.

In Phase II, these knowledge modules are deepened and transferred to practical applications, usually immediately afterwards, through exercises or case studies to be worked on by the students themselves. The results are presented by the students and then discussed. This means that in the course the basic knowledge is not only conveyed but is also applied and deepened through application examples and exercises.

In phase III the students deepen the acquired knowledge once again independently based on the literature. Continuous collaboration and reworking of the material is therefore an indispensable prerequisite for learning success. At the same time the effort for the exam preparation is minimized and the time and effort for the whole exam preparation is semester. Active participation in teaching is an elementary component of the teaching and learning concept. The teacher is always available as a discussion partner and provides support and advice. The communication takes place in personal counselling or via e-mail.

Literature and Course Materials

To be provided in class

Assessment

PLH/PLL/PLK/PLP/PLR : A written exam (60 min) or homework will be provided.

Students are expected to be present at all classes. Absences have to be excused in advance and agreed upon by the lecturer. Unexcused absence during more than two classes leads to failure of the entire course.

The grading is as follows:

- 1.0 Very good, a performance significantly above the average
- 2.0 Good, an above average performance
- 3.0 Satisfactory, an average performance
- 4.0 Adequate, a below average performance with noticeable shortcomings
- 5.0 Fail, an unacceptable performance

Schedule

1	08.06.2026 09:45 - 11:15	Drivers of the new mobility paradigm
2	08.06.2026 11:30 - 13:00	Options for the new individual mobility
3	15.06.2026 09:45 - 11:15	Options for the new individual mobility
4	15.06.2026 11:30 - 13:00	Options for the new individual mobility
5	18.06.2026 09:45 - 11:15	Electric vehicles: operating modes and new value chain (e.g. electric motor with permanent magnets or electromagnets, power electronics, battery, infrastructure etc.)
6	18.06.2026 11:30 - 13:00	Electric vehicles: operating modes and new value chain (e.g. electric motor with permanent magnets or electromagnets, power electronics, battery, infrastructure etc.)
7	18.06.2026 13:45 - 15:15	Electric vehicles: operating modes and new value chain (e.g. electric motor with permanent magnets or electromagnets, power electronics, battery, infrastructure etc.)
8	18.06.2026 15:30 - 17:00	Research Report: KIT new solid state + sodium batteries, Dr. Manuel Baumann: Coordinator of the European Energy Research Alliance, Joint Programme Energy storage, Sub-programme 6 Sustainability and techno-Economics

9	18.06.2026 17:15 - 18:45	Research Report: KIT new solid state + natrium batteries, Dr. Manuel Baumann: Coordinator of the European Energy Research Alliance, Joint Programme Energy storage, Sub-programme 6 Sustainability and techno-Economics
10	19.06.2026 09:45 - 11:15	Automated vehicles: levels of automation, sense (perceive relevant static objects, determine the location of AV, predict the future behaviour of relevant actors)
11	19.06.2026 11:30 - 13:00	Automated vehicles: Plan (create a collision free and lawful driving plan); Act (correctly execute and actuate the driving plan)
12	19.06.2026 13:45 - 15:15	Project and simulation reports: EV and new mobility services and value-added services (V2G)
13	19.06.2026 15:30 - 17:00	Project and simulation reports: Car-sharing project in Baden-Württemberg
14	19.06.2026 17:15 - 18:45	Project and simulation reports: Automated EV as mobility gap filler in a MaaS (mobility as a Service) or an ITS (Intelligent Transport System): the "AVENUE" (Horizon 2020: 2018-2022) and ULTIMO (Safe, Resilient Transport and Smart Mobility services for passengers and goods: 2022-2026) project (Horizon Europe)
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Code of Conduct for Students

[Link to the Code of Conduct for online Teaching](#)

Teaching Philosophy

Your education is important to me. Therefore, I would like to support you. Please feel free to contact me via email if you have any questions or problems regarding the course. I will respond as soon as possible and if necessary make an appointment with you.

I want to do my part to ensure that you achieve a successful learning progress and gain an understanding of sustainable mobility. I would like to create an interesting and varied lecture for you, which will pave the way for the world of work. Your comments which serve the learning progress of all are welcome. My goal is that you are successful in the lecture, but the majority of the work you must perform yourself.

Additional Information

Language: German